

# Siddharth Deepak

Leander, TX • (737) 368-2120 • [sid.deepak2023@gmail.com](mailto:sid.deepak2023@gmail.com) • [linkedin.com/in/siddharth-deepak-3182b8219](https://www.linkedin.com/in/siddharth-deepak-3182b8219) • [github.com/siddeepak2023](https://github.com/siddeepak2023) • portfolio: [siddharthdeepak.bizfalconai.com](https://siddharthdeepak.bizfalconai.com)

## EDUCATION

**B.S. in Business Analytics and AI | Concentration in Marketing and AI** Anticipated Graduation: Dec 2027  
*The University of Texas at Dallas – Naveen Jindal School of Management, Richardson, TX*

## RELEVANT COURSEWORK

Data Science for Business Applications, AI in Business, Business Analytics, Marketing Analytics, Foundations of Business Intelligence, Database Analytics, Object-Oriented Programming in Python, Quantitative Business Analysis

## TECHNICAL SKILLS

|                   |                                                                                    |                   |                                                                                                                                                         |
|-------------------|------------------------------------------------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b> | Python, SQL, TypeScript, HTML/JS/CSS (familiar with R)                             | <b>Libraries:</b> | Pandas, NumPy, scikit-learn, statsmodels, HistGradientBoosting, Optuna                                                                                  |
| <b>Tools:</b>     | Tableau, Excel, Git, SQLite, Three.js, Chart.js, Firebase (familiar with Power BI) | <b>Concepts:</b>  | ML Modeling, Model Validation & Calibration, A/B Testing, Statistical Hypothesis Testing, Feature Ablation, Time-Series Forecasting, EDA, Data Cleaning |

## PROFESSIONAL EXPERIENCE

**BizFalcon AI** [bizfalconai.com](https://bizfalconai.com) — Founder & Analytics Lead May 2026 – Present

- Shipped two analytics products: **ProfitFalcon** for e-commerce brands — contribution margin, LTV/CAC, discount leakage and stockout risk — and **ClientFalcon** for marketing agencies, multi-client reporting built on an explainable account-health score; both run on standard platform exports (Shopify, Meta, Google Ads, GA4, Klaviyo)
- Built the shared analysis engine underneath, **Talon**, which runs Python over a client's entire dataset rather than a sample; authored **9 industry metric packs** and **11 guided statistical workflows** (hypothesis testing, regression, cohort, time-series forecasting) with the formula behind every figure exposed so any number can be audited
- Designed a recommendation layer that reads the data and names the next action — cut this budget, reorder this SKU, review this price — with **projected dollar impact**, human approval required before anything is applied, then scores each decision against the realized result so recommendation quality is measurable
- Engineered and operate the full stack solo — React/TypeScript, Firebase, in-browser Python (Pyodide), 9-model LLM integration, Stripe billing — with **400+ automated tests** and CI

**YMCA Central Texas** — Summer Camp Counselor May 2021 – Aug 2025

- Led groups of 15–25 campers across four summers and trained incoming counselors through a structured shadowing program
- Awarded Counselor of the Week (5×) and Rookie Counselor of the Year for leadership and camper experience

## PROJECTS

**Retail Customer Intelligence Platform** [Live Dashboard](#)

*Python | Scikit-learn | Pandas | RFM | Market-Basket Analysis*

- End-to-end customer analytics on **802,712 real transactions** (41 countries, £17.45M): K-means RFM segmentation into 4 tiers; a random-forest churn model scored **out-of-sample at 0.796 ROC-AUC against a 55.3% base rate**; sized the retention campaign with a **precision@k curve** (top 100 contacts are 90% real churners) and a cost model showing contact capacity, not cost, is the binding constraint; Apriori market-basket rules (top pair lift 24×) powering a cross-sell engine

**NBA Game Outcome Prediction Model** [GitHub](#)

*Python | HistGradientBoosting | Optuna | SQLite | nba\_api*

- Trained a gradient-boosted classifier on **10,624 real NBA games** (9 seasons via nba\_api) and validated on the fully held-out 2024-25 season: **66.4% accuracy against a 54.4% always-home baseline** (AUC 0.727, Brier 0.211), with isotonic-calibrated probabilities, TimeSeriesSplit, and 2020 bubble games down-weighted after measuring the home-court collapse
- Ran a **feature ablation** on the 71-feature set: Elo differential alone reaches AUC 0.710, so the other 70 add only **+0.015 AUC — under the ±0.021 spread across CV folds**, improving calibration but not ranking
- Re-tested on the **complete 2025-26 season** (1,230 unseen games, independent source): AUC held at **0.718** but did not beat a closed-form Elo formula (0.718 vs 0.718), winning only on calibration (**Brier 0.212 vs 0.215**) — the value is probability quality, not ranking

**Olist E-Commerce Revenue & Sales Analysis** [Live Dashboard](#)

*Python | SQL | Pandas | Tableau | Excel*

- Analyzed **96,478 e-commerce orders** across 9 source tables, joined into a star schema and queried in SQLite with CTEs and window functions; Health & Beauty led revenue at R\$1.4M of R\$15.4M. At order grain, a logistic regression controlling for state, category and freight put a one-week delivery slip at **1.78× the odds of a ≤3-star review** (OR 1.085/day, 95% CI 1.083–1.088, n=95,824) — a larger effect than the raw 3.01★ vs 4.41★ gap implies. Published as a live Tableau dashboard.